



UKUDLA Data Science Certificate

Data Integration

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UKUDLA
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Systems and Applied Agricultural
and Food Data Science

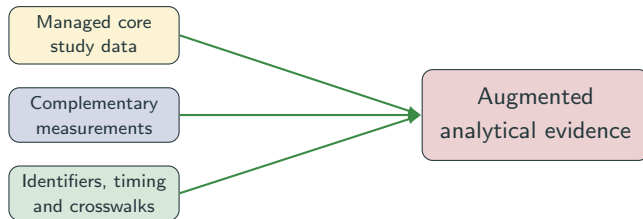


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Table of contents

1. Motivation
2. Concepts
3. Application
4. Key Message

A managed dataset may not answer the question



One managed source rarely contains every outcome, exposure, treatment, covariate or context required by a study. Additional measurements or datasets can add evidence—if their observations can be aligned responsibly.

Data integration combines information from different sources into a coherent unit of analysis while preserving meaning and lineage.

Data integration extends data management

Manage each source

Describe, validate and preserve both the existing and additional data.

Integrate deliberately

Align identifiers, grains, periods, spaces, units and classifications.

Manage the result

Audit the join and document the new artifact, assumptions and lineage.

This session moves from **motivation**, through the essential **concepts**, to an auditable **application** across compatible sources.

Data acquisition supports this process; integration is the teaching focus.

Sources bring different representations

Core study data

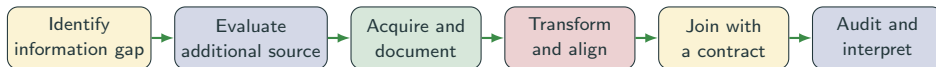
- laboratory samples or assay results;
- experimental plots and treatments;
- observational units and visits; or
- entities and periods from secondary data.

Complementary data

- instrument, sensor or laboratory measurements;
- management, survey or administrative records;
- weather, soil or remote-sensing products; and
- different grains may require aggregation.

Different representations must be made compatible without hiding how their meaning, space and time differ.

A data need initiates integration



The research question determines what information is missing, which source is appropriate, and what the integrated observation should represent.

Do not acquire an additional dataset before defining why and how it will be integrated.

Compatibility has several dimensions

Structural compatibility

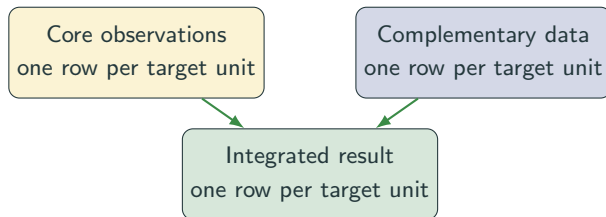
- identifiers and candidate keys;
- dataset grain and relationships;
- variable types and units;
- missing-value conventions; and
- classifications and code lists.

Scientific compatibility

- population and geographic support;
- observation and reporting periods;
- measurement definitions;
- aggregation assumptions; and
- fitness for the intended question.

A technically successful join does not prove that the combined variables refer to comparable evidence.

State the grain before joining



- A one-to-one join should preserve the expected key and row count.
- A one-to-many relationship can legitimately multiply rows.
- An unexpected many-to-many relationship often indicates an incomplete key.
- The join function cannot decide whether the scientific relationship is valid.

Stable identifiers and crosswalks connect sources

Names are fragile

- spelling and punctuation differ;
- abbreviations differ;
- names and boundaries change; and
- the same name can identify different entities.

Source label	Stable ID
Plot 1	SITE-A-P001
Sample 17	LAB-2026-0017
Farm North	FARM-0042

A crosswalk is a documented dataset, not an invisible correction embedded in code.

Check unmatched keys, duplicated keys, new missing values and row multiplication after every join.

Integration requires alignment, not only a join

Dimension	Possible mismatch	Decision to document
Identifiers	Sample, plot, farm and provider codes	Crosswalk and valid period
Time	Assay, visit, season or reporting period	Alignment and period definition
Space	Plot, site, region, point or raster cell	Boundary, CRS, extent and summary rule
Units	Mass, area, concentration, rate or index	Conversion and dimensional consistency
Schema	Names, types, categories and flags	Mapping and retained information

Aggregating replicates, visits, sensor readings or gridded values is an analytical choice, not a neutral format conversion.

Validate the result and preserve lineage

Join audit

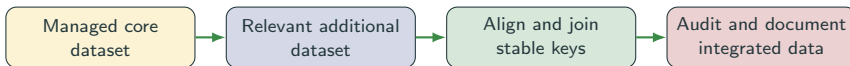
- expected and actual row counts;
- matched and unmatched keys;
- duplicated or multiplied records;
- missingness introduced; and
- resulting grain and coverage.

Lineage record

- source of every retained variable;
- crosswalks and conversion rules;
- temporal or spatial aggregation;
- script and input versions; and
- known losses and limitations.

A critical failed expectation should stop the pipeline visibly rather than produce a plausible-looking result silently.

Key message: integration requires explicit alignment



1. Identify the additional information required by the question.
2. Reconcile grain, identifiers, time, space, definitions and units.
3. Map source labels to stable analytical identifiers.
4. Join on tested keys under an explicit integration contract.
5. Save the derived table, audit, dictionary and lineage record.

A successful join is not sufficient: integrated data inherit every source and alignment limitation.

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